

TrES-2b

R-1721

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_150725 · created 2026-08-02T16:33:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457228.72709 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.091 +/- 0.042
Transit depth (Rp/Rs)^2	0.83 +/- 0.76 [%]
Orbital Inclination (inc)	79.1 +/- 2.96
Airmass coefficient 1 (a1)	1.41 +/- 0.18
Airmass coefficient 2 (a2)	-0.32 +/- 0.4
Scatter in the residuals of the lightcurve fit is	12.56 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.009 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.091 $\pm$ 0.042 vs 0.1254 (deviation 0.67 σ)
Duration fit vs published	0.009 d vs 0.0746 d (deviation 2.44 σ)
Mid-time O-C	-0.8 min
Depth SNR	1.8
Residual scatter	12.56 %

Light curve

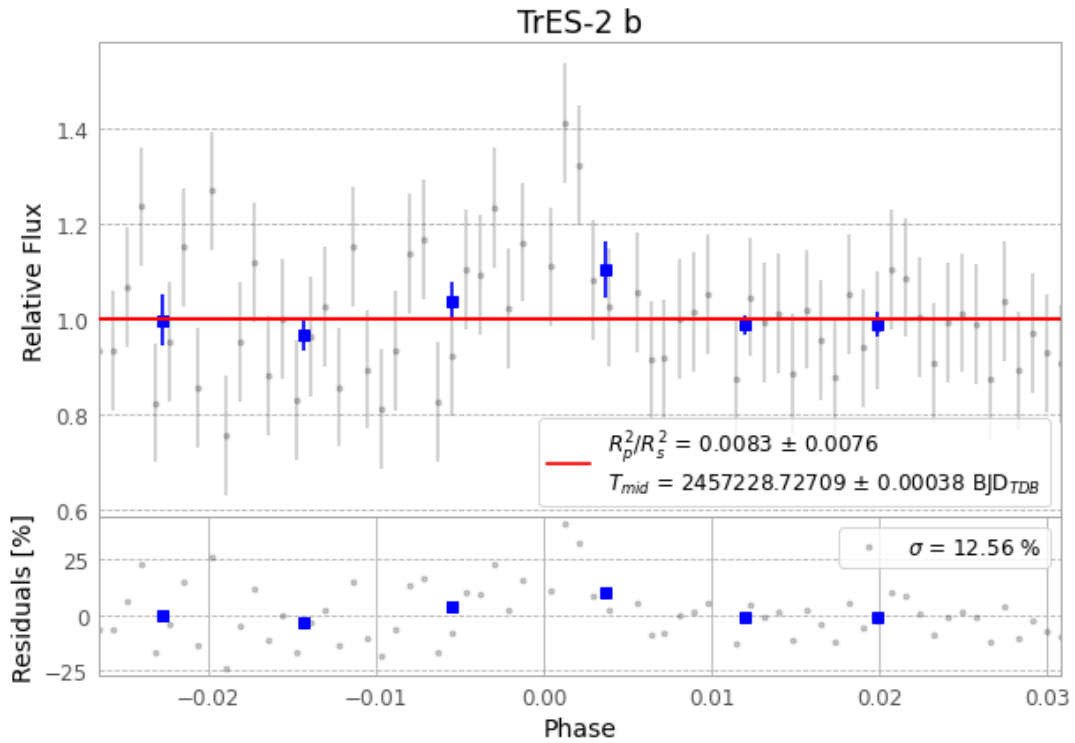


Figure 1 — FinalLightCurve_TrES-2 b_2015-07-24.png (SHA-256 80331a450dc53a6c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3a7de32e1c88f15f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), TRES-2150725034812.FITS → TRES-2150725071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-150725.log (2f5ad6d4f225...), FinalLightCurve_TrES-2 b_2015-07-24.png (80331a450dc5...), FinalLightCurve_TrES-2 b_2015-07-24.csv (9e68cd9136bd...)

Compute resources

Wall time 240.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:33Z.